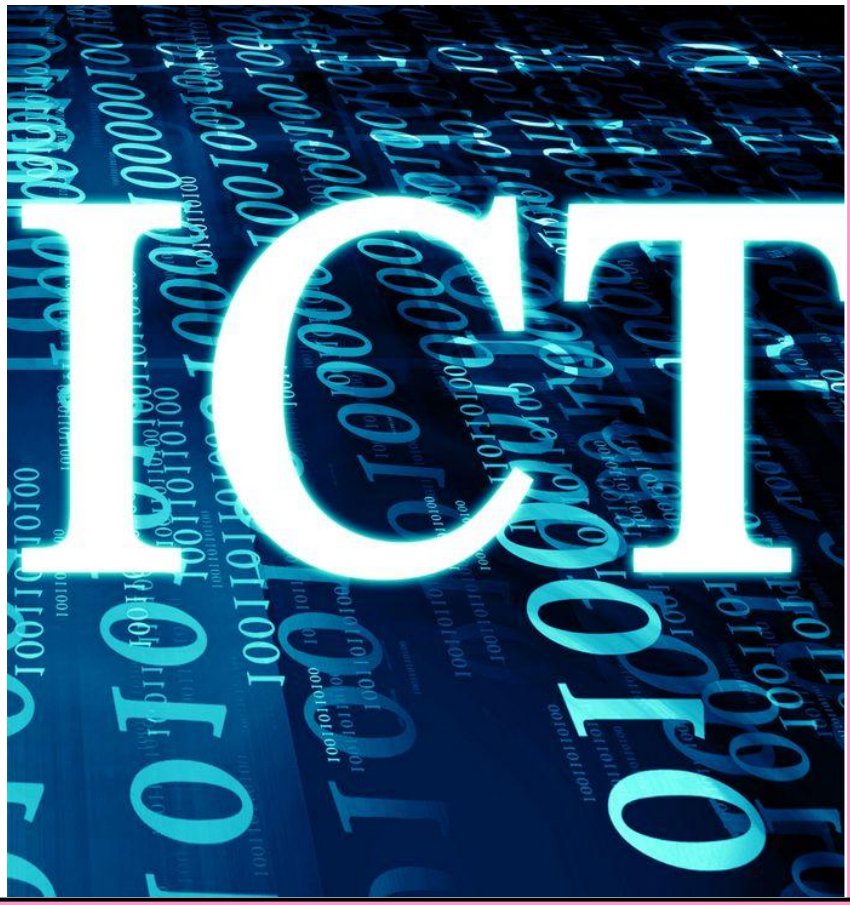


Work Integrated Learning- XITM6329

DITM - GROUP 3

Tlotlang Ramatlotlo ST10445028

Praise Kapinga ST10452822

Zandile Nkomo ST10440938

Nomnotho Nshangase ST10462439

Kyle Bila ST10452906

Development of selection criteria

The project was selected using different criteria to decide if it is suitable (Gido, Clements & Baker, 2018). One of the most important criteria is alignment with goals, because the project focuses on improving healthcare services in the Sekhukhune District by improving the network at the New Jane Furse Memorial Hospital. This includes improving how patient records are stored and accessed, allowing different departments in the hospital to share information faster and making it easier for the hospital to communicate with external suppliers.

Another criterion is improving operations, since the hospital currently has only 15 outdated computers, 3 printers and no proper security systems. There are also issues with missing patient files and poor data storage, which affects how the hospital provides services. The project plans to fix these problems by introducing a more reliable and secure network system. The project also considers cost-effectiveness, because it uses the current municipal network and extends it to the hospital, instead of building a completely new network from the beginning (Gido, Clements & Baker, 2018).

Lastly, long-term use is important. The hospital only has three ICT staff members who are responsible for maintaining the network. This means the system must be simple enough for them to manage, including basic tasks like troubleshooting, restarting systems and replacing damaged equipment without needing constant external support.



Figure 1: Jane Furse Memorial Hospital

Assumptions

There are a few assumptions made for this project (Gido, Clements & Baker, 2018). It is assumed that the municipality's current network infrastructure is strong and stable enough to be extended over the 1 km distance to the hospital. It is also assumed that Openserve will be available as the main service provider to support the connection in this rural area.

Another assumption is that the three ICT staff members will be successfully trained to install, manage and fix problems in the new network system. It is also assumed that hospital staff will get used to the improved system, especially when it comes to managing patient records digitally instead of relying on systems where files can go missing. Lastly, it is assumed that the project can be fully completed in one year.

Data and information gathering

Information about the hospital was collected to support the project selection (Gido, Clements & Baker, 2018). The hospital currently has only 15 very old computers and 3 printers, which are not enough to support the daily operations of a hospital that serves around 100 patients per day. The hospital also has about 600 staff members, including medical staff, administrative staff and contractors, which increases the need for a better system.

There are also serious issues with data management, such as missing patient files and storage systems that can't be depended on. This shows that the current infrastructure is not able to properly support important hospital functions like patient record management and communication between departments.

Also, information about the environment was considered. The hospital is located only 1 km from the municipality, which makes it possible to extend the current network instead of building a new one. Feedback from stakeholders such as ICT staff and hospital management is also important to understand the challenges they have and what improvements are needed in the system.

Evaluation of the project

The project was evaluated based on its benefits and challenges (Gido, Clements & Baker, 2018). Another scenario, which focuses on developing a community network in the Central Karoo district to connect 3000 homes, government buildings and hospitals was also considered as a potential project. However, this scenario would require a much larger infrastructure, including wireless main network systems, solar-powered network operating centres and coverage across a wide rural area which makes it more difficult and requires more resources to complete in one year.

In comparison, the Sekhukhune hospital project is more focused and easier to manage. It only requires extending the municipal network over a short distance to a single main hospital, which reduces complexity, cost and risk. The benefits of this project include improving patient record management, reducing the issue of missing files, improving communication between hospital departments and allowing better access to external systems used by suppliers.

The project is also more manageable because it can be maintained by the three ICT staff members after proper training. However, there are still challenges such as relying on one service provider and ensuring that staff are properly trained.

Overall, the project provides both clear benefits such as improved data storage, better communication systems and easier access to patient information and long-term benefits such as improved healthcare service delivery in the district. Based on this evaluation, the Sekhukhune hospital network project is the most suitable option to be implemented.

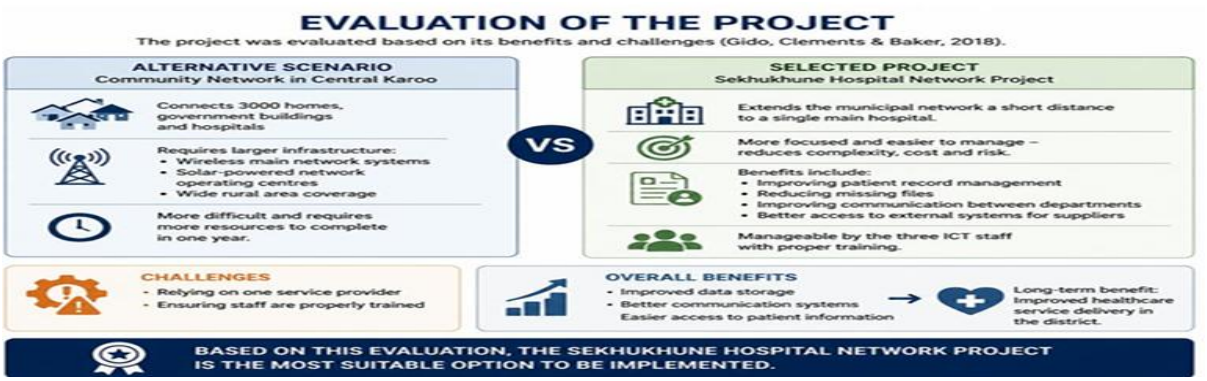


Figure 2: Evaluation of the Project

Executive summary

The Sekhukhune District Municipality has identified the need to improve digital connectivity between the municipality and New Jane Furse Memorial Hospital to strengthen healthcare service delivery and reduce the digital divide between rural and urban communities. The hospital currently operates with outdated information technology infrastructure, limited cybersecurity measures, insufficient data storage, and inadequate network capacity. These challenges affect patient record management, communication with suppliers, administrative efficiency, and the overall quality of healthcare services.

This project proposal presents a comprehensive and sustainable network infrastructure solution that will connect the hospital to the existing municipal network through a dedicated Openserve fibre connection. The proposed project includes fibre deployment, structured cabling, local area network implementation, wireless connectivity, server and storage infrastructure, cybersecurity controls, backup and disaster recovery systems, and ICT staff training. The project will be implemented over a period of 12 months and is designed to provide reliable, secure, and scalable network services.

The proposal is developed using the principles of project management outlined by (Gido, Clements & Baker, 2018), particularly the importance of project planning, stakeholder management, scope definition, scheduling, budgeting, risk management, and sustainability.

Introduction

Modern healthcare systems rely heavily on digital technologies to support patient care, administrative functions, financial management, procurement, and communication with external stakeholders. A well-designed network infrastructure enables hospitals to access electronic medical records, coordinate with suppliers, secure sensitive information, and improve operational efficiency. In rural areas, however, limited connectivity and inadequate ICT

infrastructure often undermine service delivery and create barriers to effective healthcare management.

New Jane Furse Memorial Hospital serves as the main public healthcare facility for the Sekhukhune District in Limpopo Province. The hospital provides essential healthcare services to a large rural population, yet its existing ICT infrastructure is insufficient to support modern healthcare operations. The current network environment is characterised by outdated computers, limited printing facilities, inadequate security measures, and concerns regarding missing patient records and data storage.

The purpose of this project proposal is to present a sustainable network infrastructure solution that extends the existing municipal network to the hospital through a dedicated fibre connection. The proposal also addresses cybersecurity, data management, staff capacity development, and long-term operational sustainability. According to (Gido, Clements & Baker, 2018), successful project management requires a structured approach that integrates technical planning with stakeholder engagement and organisational objectives. This proposal follows that approach to ensure successful implementation within the required one-year timeframe.

Project background

The Sekhukhune District Municipality consists of four local municipalities: Elias Motsoaledi, Ephraim Mogale, Fetakgomo Tubatse, and Makhuduthamaga. As part of its digital development strategy, the municipality intends to improve connectivity between municipal facilities and critical public institutions. New Jane Furse Memorial Hospital has been identified as a priority site because it functions as the largest public sector hospital within the district and provides comprehensive healthcare services to the surrounding communities.

The hospital employs approximately 600 staff members, including medical personnel, administrative staff, and contractors, while serving approximately 100 patients per day. Despite its importance, the hospital currently has only 15 old computers and three printers, which are inadequate for an institution of this size. Furthermore, only three ICT support staff are responsible for managing the entire network environment.

The hospital is located approximately 1 kilometre from the municipality, making it technically feasible to extend the existing municipal fibre network to the hospital. Openserve has been identified as the most appropriate carrier for the connection, providing an opportunity to establish a dedicated and reliable communication link between the two facilities.

Problem statement

The existing ICT infrastructure at New Jane Furse Memorial Hospital is inadequate to support efficient healthcare service delivery. The outdated computer systems, limited network capacity, insufficient security controls, and poor data management practices have created operational inefficiencies that affect both clinical and administrative services.

One of the most significant concerns is the management of patient information. Missing patient files and inadequate data storage systems compromise healthcare continuity and administrative efficiency. The hospital also relies on communication with suppliers through Intralink services, making reliable network connectivity essential for procurement and supply chain operations. In addition, the absence of adequate cybersecurity measures exposes sensitive healthcare information to potential cyber threats, unauthorised access, and data breaches.

Without a comprehensive network infrastructure upgrade, the hospital will continue to face increasing operational risks, reduced service quality, and limitations in future digital healthcare initiatives.

Project objectives

The main objective of the project is to design and implement a secure, reliable, and sustainable network infrastructure that connects New Jane Furse Memorial Hospital to the Sekhukhune municipal network within 12 months.

The project aims to establish a dedicated fibre connection between the municipality and the hospital, modernise the hospital's network infrastructure, improve data management and storage, implement comprehensive cybersecurity controls, provide secure wireless

connectivity, enhance communication with suppliers, and develop the technical capacity of the hospital's ICT support staff. (Gido, Clements & Baker, 2018) emphasise that clearly defined project objectives provide the foundation for effective planning, monitoring, and project success.

Project scope

The project scope defines the work required to deliver the proposed network infrastructure solution. The implementation will include fibre network extension, structured cabling throughout the hospital, installation of core networking equipment, deployment of enterprise wireless infrastructure, installation of servers and storage systems, implementation of cybersecurity controls, backup and disaster recovery systems, staff training, system testing, documentation, and final project handover.

The project will provide network connectivity to 20 patient wards and 5 administrative offices, while also supporting finance, supply chain, pharmacy, management, and records departments.

The project does not include construction of new hospital buildings, procurement of clinical medical equipment, recruitment of additional permanent ICT staff, or development of new hospital information systems. (Gido, Clements & Baker, 2018) note that a clearly defined project scope reduces uncertainty and helps prevent scope creep during project implementation.

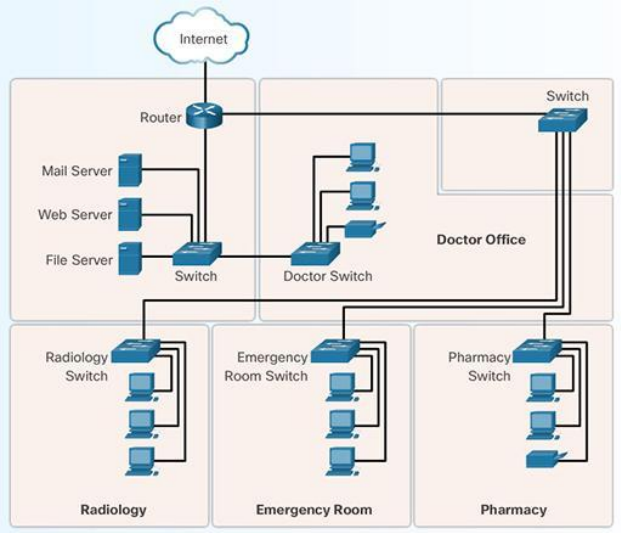


Figure 3: Local area network connecting hospital wards and administrative offices.

Stakeholder analysis

Stakeholder engagement is particularly important because the project involves both technical infrastructure and organisational change. The municipality, hospital management, Openserve, ICT staff, healthcare workers, suppliers, and patients all have an interest in the project outcome.

The Sekhukhune District Municipality functions as the project sponsor and owner of the municipal network infrastructure. Hospital management serves as the primary client responsible for operational implementation and institutional support. Openserve provides the fibre connectivity infrastructure and technical coordination. The ICT support staff will operate and maintain the new network infrastructure, while medical and administrative personnel will be the primary users of the system.

Effective communication among stakeholders is essential for project success. (Gido, Clements & Baker, 2018) explain that stakeholder involvement throughout the project life cycle improves acceptance, reduces resistance to change, and supports long-term sustainability.

Proposed network infrastructure solution

The proposed solution is based on a fibre-to-the-site architecture that extends the existing municipal network directly to New Jane Furse Memorial Hospital. Fibre optic technology has been selected because it provides high bandwidth, low latency, excellent reliability, and long-term scalability.

A dedicated Openserve fibre connection will be installed between the municipality and the hospital over the existing 1 km route. The fibre connection will terminate at the hospital's main server room, where a secure core network infrastructure will be established.

The hospital network will follow a hierarchical design consisting of a core layer, distribution layer, and access layer. The core layer will include enterprise-grade routers, Layer 3 switches, and a next-generation firewall. The distribution layer will connect the various hospital departments,

while the access layer will provide connectivity to end-user devices, printers, and wireless access points.

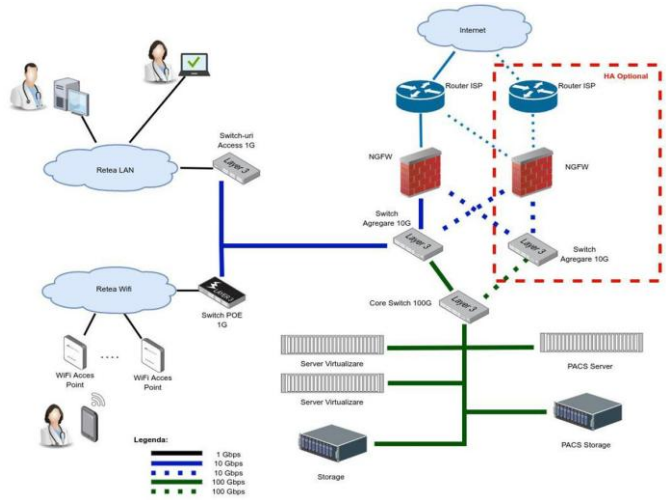


Figure 4: Proposed network topology for New Jane Furse Memorial Hospital.

Network segmentation will be implemented through virtual local area networks (VLANs) to separate clinical systems, administrative systems, supplier access, guest wireless services, and network management traffic. This approach improves security, network performance, and administrative control.



Figure 5: Fibre-optic connectivity between the municipality and the hospital.

Hardware and software infrastructure

The project proposes the replacement of outdated ICT equipment with modern enterprise infrastructure. Approximately 80 desktop computers, 10 laptops, and 15 network printers will be installed throughout the hospital to support clinical and administrative operations.

The server environment will include application servers, database servers, file servers, backup servers, and network attached storage systems. These servers will support electronic patient records, financial systems, supply chain management, document storage, and network management services.

The software environment will include Windows Server, Active Directory, Microsoft SQL Server, enterprise backup software, endpoint security software, network monitoring tools, and centralised authentication services. These technologies will provide improved reliability, user management, and operational efficiency.

Cybersecurity strategy

Cybersecurity is a critical component of the proposed network infrastructure because healthcare institutions manage highly sensitive personal and medical information. The current absence of adequate security measures exposes the hospital to significant cyber risks, including ransomware attacks, unauthorised access, data breaches, and service disruptions.

The proposed solution includes the deployment of a next-generation firewall, intrusion detection and prevention systems, endpoint protection, multi-factor authentication, encrypted communications, role-based access control, automated patch management, security monitoring, and regular vulnerability assessments.

Security policies will also be implemented to regulate user access, password management, supplier connectivity, remote access, and data protection. Staff awareness training will be conducted to reduce the risk of phishing attacks, social engineering, and other human-related security threats.

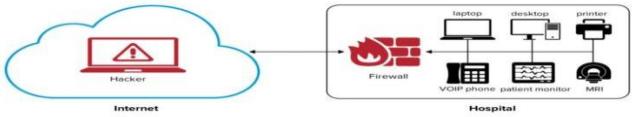


Figure 6: Cybersecurity architecture for the proposed hospital network.

Data management and disaster recovery

The hospital currently experiences challenges related to missing patient files and inadequate data storage. To address these issues, the project will implement a centralised electronic data management system supported by secure server infrastructure.

Daily automated backups will be performed, with copies stored both on-site and off-site to ensure data availability during disasters. Disaster recovery procedures will be developed to support rapid restoration of critical systems in the event of equipment failure, cyber incidents, or natural disasters.

The implementation of reliable backup and recovery systems will significantly improve information availability, operational continuity, and regulatory compliance.

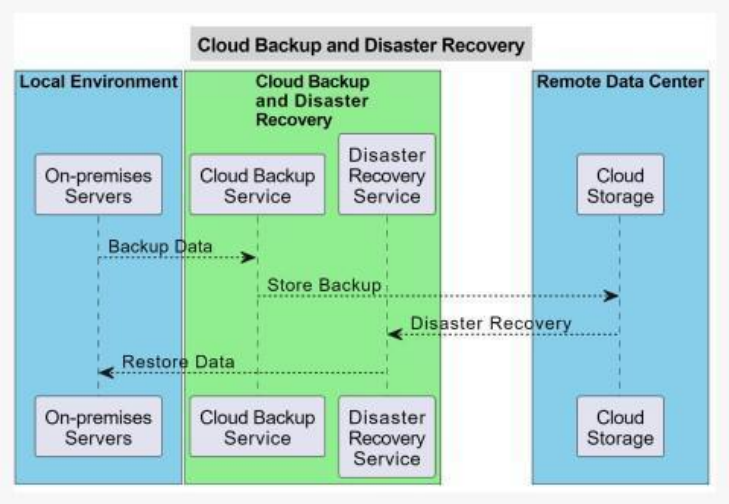


Figure 7: Backup and disaster recovery process for hospital information systems.

Project schedule

The project will be implemented over a period of 12 months. The first month will focus on project initiation, stakeholder approval, and establishment of the project team. Detailed network design and procurement activities will be completed during the second and third months.

Fibre installation will be undertaken during months four and five, followed by local area network deployment and structured cabling during months six and seven. Server installation, cybersecurity implementation, and system integration will occur during months eight and nine. Comprehensive testing and quality assurance will be conducted during month ten.

Training of the three ICT support staff will take place during month eleven, while final documentation, system handover, and project closure will be completed during month twelve. (Gido, Clements & Baker, 2018) emphasise that realistic scheduling provides the framework for coordinating activities, managing resources, and monitoring project progress.

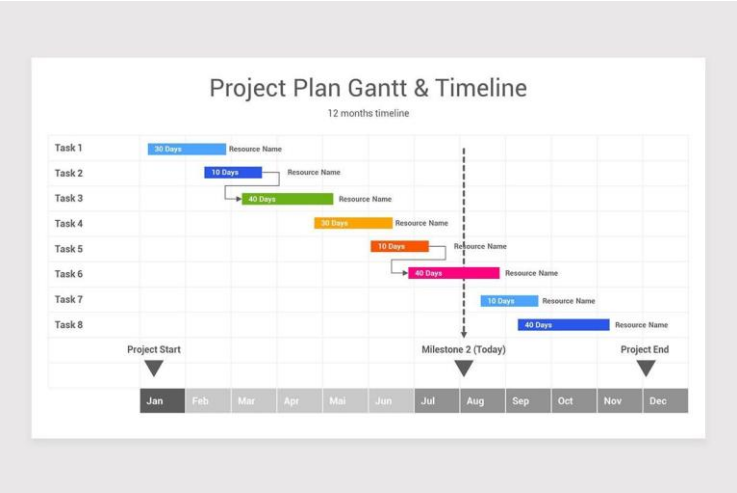


Figure 8: Proposed 12-month implementation timeline.

Budget estimate

The estimated total project budget is R5,335,000. Fibre installation is estimated at R1,200,000, while network equipment, including routers, switches, firewalls, and wireless infrastructure, is estimated at R850,000. Computer and printer replacement is estimated at R900,000, and server and storage infrastructure is estimated at R750,000.

Cybersecurity systems are budgeted at R500,000, structured cabling and installation at R400,000, ICT staff training at R150,000, and testing and documentation at R100,000. A contingency reserve of R485,000 has been included to accommodate unforeseen costs and

project risks. According to (Gido, Clements & Baker, 2018), contingency planning is an important aspect of project budgeting because it improves financial control and project resilience.

Risk management

Several risks may affect the successful implementation of the project. Delays in fibre installation could affect the project schedule, while procurement delays may postpone equipment deployment. Budget overruns, cybersecurity incidents during implementation, ICT skills shortages, resistance to organisational change, and power interruptions also represent significant project risks.

These risks will be managed through early coordination with Openserve, strict procurement planning, regular budget monitoring, implementation of cybersecurity controls, comprehensive staff training, stakeholder communication, and installation of uninterruptible power supply systems. (Gido, Clements & Baker, 2018) describe risk management as a continuous process that involves identifying risks, assessing their impact, developing response strategies, and monitoring their occurrence throughout the project life cycle.

Training and sustainability

The sustainability of the network infrastructure depends largely on the technical capacity of the hospital's ICT support staff. Because only three ICT personnel currently manage the hospital network, comprehensive technical training is essential.

The training programme will cover fibre equipment installation, router and switch configuration, firewall management, wireless administration, backup procedures, equipment replacement, troubleshooting, network monitoring, and cybersecurity incident response. The training will combine classroom instruction with practical hands-on exercises.

Long-term sustainability will also be supported through maintenance agreements with Openserve, preventive maintenance schedules, documented operating procedures, annual

security assessments, and scalable network architecture that can accommodate future expansion.



Figure 9: ICT staff receiving network installation and maintenance training

Monitoring and evaluation

Project performance will be monitored throughout implementation using measurable performance indicators. These indicators include network availability above 99.5%, successful completion within 12 months, reduction in missing patient files, successful completion of ICT staff training, reduction in network incidents, improved user satisfaction, and compliance with cybersecurity standards.

Regular progress reports will be submitted to the municipality and hospital management to ensure effective governance and timely corrective action where necessary. (Gido, Clements & Baker, 2018) emphasise that continuous monitoring and evaluation enable project managers to compare actual performance with planned objectives and maintain project control.

Conclusion

The proposed network infrastructure project provides a comprehensive and sustainable solution for improving digital connectivity between the Sekhukhune District Municipality and New Jane Furse Memorial Hospital. By extending the municipal fibre network to the hospital and implementing modern networking, cybersecurity, and data management infrastructure, the

project will significantly improve healthcare service delivery, information security, operational efficiency, and long-term organisational capacity.

The proposal aligns with recognised project management principles by providing clearly defined objectives, scope, stakeholder responsibilities, scheduling, budgeting, risk management, and sustainability strategies. Successful implementation of the project will create a reliable technological foundation for healthcare delivery within the Sekhukhune District and support future digital transformation initiatives across the municipality.

PROJECT PLAN: NEW JANE FURSE MEMORIAL HOSPITAL ICT INFRASTRUCTURE UPGRADE

EXECUTIVE SUMMARY

This project plan details the strategy for designing, deploying, and commissioning a dedicated network infrastructure connecting the Sekhukhune District Municipality to the New Jane Furse Memorial Hospital in Limpopo, South Africa. The project addresses critical operational vulnerabilities, including missing patient records, lack of cybersecurity controls, outdated hardware, and limited connectivity.

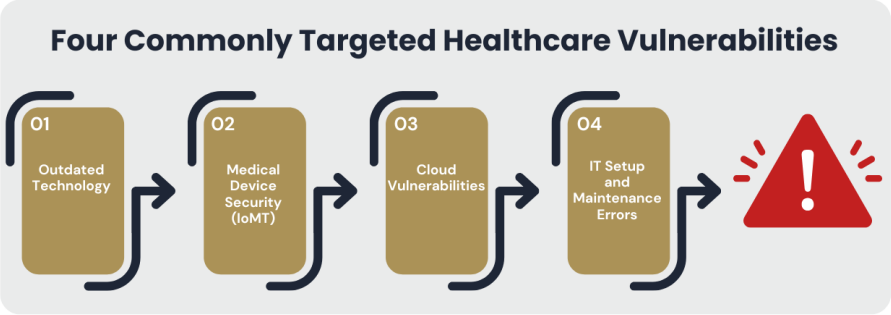


Figure 10: Four commonly Targeted Healthcare Vulnerabilities

By splicing off the municipality's core network located 1 km away, the project will deliver a dedicated, high-speed fibre link to the hospital, upgrade core networking equipment, implement robust cybersecurity protocols, and upskill the local IT workforce. This will ensure that the hospital can rapidly benefit from improved connectivity, data management, and operational efficiency.

The total project lifecycle span is 12 months, operated under a fixed-scope approach to ensure a self-sustaining system upon rollout. By training the existing three ICT support staff, the project guarantees long-term sustainability, reducing the hospital's dependence on external technical

support. Ultimately, this initiative represents a significant step towards closing the digital divide in rural South Africa and improving the quality of life for the people of Sekhukhune District.

PROJECT PURPOSE

The purpose of this project is to cover the digital divide between rural and urban coverage in the Sekhukhune District by extending reliable municipal ICT infrastructure to the district's primary healthcare facility. Establishing a secure, high-speed connection will resolve chronic data storage issues, prevent the loss of patient records, secure operational, financial, and supply chain systems, and enable real-time communication with critical healthcare suppliers.

PROJECT OBJECTIVES

Guided by the project lifecycle framework outlined in Gido et al., the project aims to fulfil the following specific objectives within a 12-month time frame:



Figure 11: Project Objectives

- Infrastructure Linkage: Splicing from the Sekhukhune District Municipality network and laying a 1 km dedicated fibre-optic connection to New Jane Furse Memorial Hospital using Openserve as the primary carrier.

- Network Security & Hardware Modernisation: Deploy firewall appliances, intrusion prevention systems, secure routing and switching hardware, and replace 15 legacy workstations and 3 outdated printers across 20 wards and 5 administrative offices. The new equipment will be more energy-efficient, reducing the hospital's operational costs and environmental footprint.

- Capacity Building: Deliver technical training to all 3 resident hospital ICT support personnel covering equipment installation, routine maintenance, rebooting procedures, and hardware unit replacement. ICT staff will also receive training on how to replace failed units and escalate issues when necessary.

- System Sustainability: Ensure seamless data storage and access to Intralinks services for suppliers while leaving behind a fully self-sustaining network environment (Gido, Clements & Baker, 2018).

STAKEHOLDERS

In accordance with PMBOK stakeholder management principles, key project stakeholders include:

- Project Sponsor: Sekhukhune District Municipality, comprising Elias Motsoaledi, Ephraim Mogale, Fetakgomo Tubatse, and Makhuduthamaga local municipalities.

- Client and End-User Institution: New Jane Furse Memorial Hospital management, 600 medical and administrative staff, and approximately 100 daily patients.

- Primary Telecommunications Carrier: Openserve.

- Local Operations Team: 3 Hospital ICT Support Staff.

- External Stakeholders and Partners: Intralinks service providers, medical suppliers, and local community representatives.

PROJECT SCHEDULE

The overall project duration is 12 months.

	ACTIVITY	RESPON.	DUR. ESTIM.	EARLIEST START	EARLIEST FINISH	LATEST START	LATEST FINISH	SLACKS
1	Project charter approval	PM	0.5	0	0.5	0	0.5	0
2	Status Reporting	PM	11.5	0.5	12	0.5	12	0
3	Design & Planning	NETENG.	1	0.5	1.5	0.5	1.5	0
4	Conduct Site Survey	NETENG.	0.5	1.5	2	1.5	2	0
5	Procurement	PROC	2	1.5	3.5	1.5	3.5	0
6	Infrastructure	SUB	2	3.5	5.5	3.5	5.5	0
7	Hardware & Software	SEC	1	5.5	6.5	5.5	6.5	0
8	Install Firewall &Configure Security	SEC	0.5	6.5	7	6.5	7	0
9	Testing and quality assurance	NETENG.	1	7	8	7	8	0
10	Training & Capacity Building	TRAINER	0.5	7	7.5	7.5	8	0.5
11	Project closure	PM	0.5	8	8.5	8	8.5	0

Figure 12: Project Schedule

SUMMARY

The project can be completed in 8.5 months.

The critical path is 1 → 3 → 5 → 6 → 7 → 8 → 9 → 11.

These 8 activities have zero slack and must be closely monitored.

Activity 10 (Training) has 0.5 months of slack and can be delayed slightly.

The project has +3.5 months of positive slack against the 12-month target, providing a buffer for unexpected delays.

RISKS & RISK MANAGEMENT PLAN



Figure 13: Project Risk & Risk Management Plan

Risk 1: Political and Social Friction

Context: Misalignment between local municipal leadership, human networks, and hospital community interests.
Impact: High impact, medium probability.
Mitigation Plan: Engage local municipal councillors early and ensure transparent communication to build social stability around the project.

Risk 2: Vendor Dependency Delays

Context: Potential delays in carrier connection activation by Openserve.
Impact: High impact, medium probability.
Mitigation Plan: Establish formal Service Level Agreements (SLAs) during planning and hold bi-weekly tracking meetings with carrier representatives (Gido, Clements & Baker, 2018).

Risk 3: Cybersecurity Vulnerabilities

Context: Cyberattacks or unauthorized intrusions during system migration.
Impact: High impact, low probability.
Mitigation Plan: Deploy firewalls and network segmentation prior to connecting internal servers to the broader network.

Risk 4: Operational Sustainability

Context: Inability of local ICT staff to maintain infrastructure post-launch.
Impact: High impact, medium probability.
Mitigation Plan: Deliver comprehensive hands-on troubleshooting and hardware replacement training to all 3 ICT staff before final sign-off.

Risk 5: Physical Cable Damage

Context: Accidental cutting or vandalism along the 1 km fibre route.
Impact: Medium impact, low probability.
Mitigation Plan: Enforce deep trenching and protective conduit casing along the entire route.

COMMUNICATION STRATEGY & PLAN

Effective communication ensures continuous stakeholder alignment and prevents project delays:

Sekhukhune District Municipality (Sponsor)

- Channel: Written Status Report & Monthly Steering Meeting.
- Frequency: Monthly.
- Purpose: Track budget, monitor milestone progress, and resolve high-level risks (Gido, Clements & Baker, 2018).

Hospital Management & Department Heads

- Channel: Operational Briefings.
- Frequency: Bi-weekly.
- Purpose: Coordinate installation across 20 wards and 5 offices without disrupting medical operations.

Openserve (Primary Carrier)

- Channel: Technical Coordination Meetings.
- Frequency: Weekly.
- Purpose: Oversee fibre link progress, municipal splicing, and wayleave approvals.

Hospital ICT Support Staff (3 Personnel)

- Channel: Stand-up Meetings & Practical Workshops.
- Frequency: Weekly during planning; Daily during hardware rollout.

- Purpose: Build system familiarity, guide installation, and conduct troubleshooting training.

Medical Suppliers & Intralinks Partners

- Channel: Electronic Circular / Bulletin.
- Frequency: Milestone-based.
- Purpose: Communicate upgrades to data storage and secure Intralinks access protocols.

SUCCESS INDICATORS

Project success will be evaluated using both quantitative and qualitative Key Performance Indicators (KPIs):

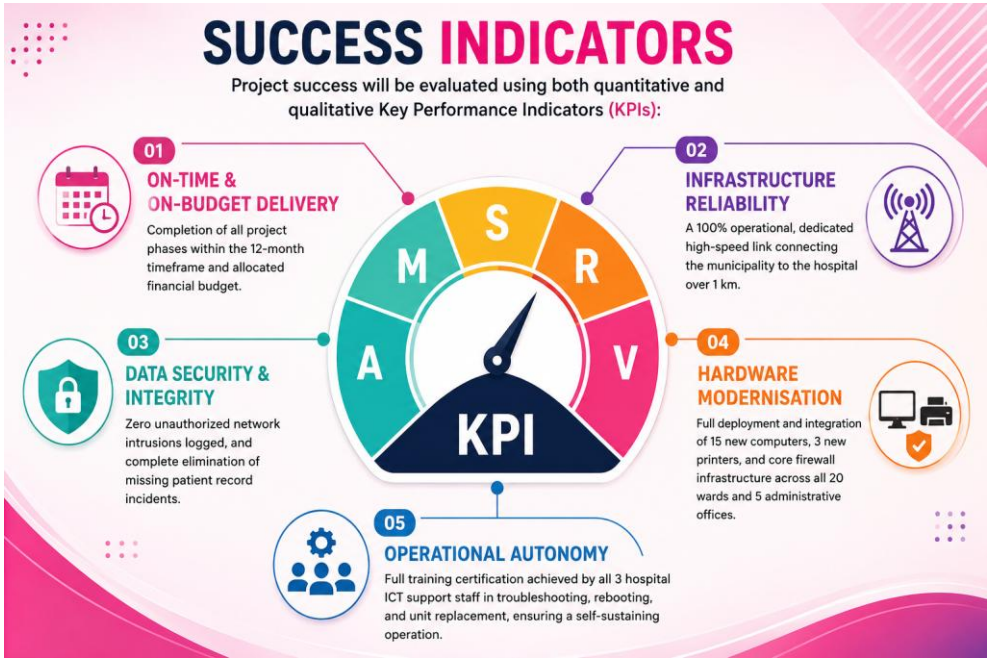


Figure 13: Success Indicators

- On-Time & On-Budget Delivery: Completion of all project phases within the 12-month timeframe and allocated financial budget.

- Infrastructure Reliability: A 100% operational, dedicated high-speed link connecting the municipality to the hospital over 1 km.

- Data Security & Integrity: Zero unauthorized network intrusions logged, and complete elimination of missing patient record incidents.

- Hardware Modernisation: Full deployment and integration of 15 new computers, 3 new printers, and core firewall infrastructure across all 20 wards and 5 administrative offices.

- Operational Autonomy: Full training certification achieved by all 3 hospital ICT support staff in troubleshooting, rebooting, and unit replacement, ensuring a self-sustaining operation (Gido, Clements & Baker, 2018).

PROJECT CHARTER

Sekhukhune District Municipality

ICT NETWORK, CYBERSECURITY & FIBRE CONNECTIVITY PROJECT

Project Sponsor	Sekhukhune District Municipality
Project Manager	
Project Classification	Municipal ICT Infrastructure Project
Version	1.0
Status	Draft for Approval
Date	10 August 2026

1. Project Charter Purpose

This Project Charter formally authorises the project and establishes its strategic purpose, governance, scope, responsibilities, authority, success measures, key assumptions, constraints and high-level delivery approach. It provides a common reference point for the Project Sponsor, Project Manager, technical specialists, procurement function, subcontractor and hospital ICT staff.

The charter is intended to support disciplined delivery of the ICT infrastructure work while protecting service continuity, information security, quality, budget and stakeholder interests.

2. Executive Summary

The project will plan, coordinate, implement, test and hand over an ICT connectivity solution involving network engineering, security controls, procurement, trenching and fibre termination, together with integration into the existing environment and training of three hospital ICT staff members.

The project structure places the Sekhukhune District Municipality as Project Sponsor, with a Project Manager responsible for day-to-day coordination. The Network Engineer, Security Specialist and Procurement Officer provide specialist leadership, while the Subcontractor performs trenching and fibre termination activities. Hospital ICT Staff 3 participate in training, operational readiness and legacy integration.

The project should be executed through controlled phases: initiation, requirements and site assessment, design, procurement, civil/fibre works, network/security implementation, integration and testing, training, handover and closure.

3. Project Objectives

- Deliver a reliable and fit-for-purpose ICT connectivity solution aligned with municipal and hospital operational requirements.
- Provide the required network infrastructure and fibre connectivity, including trenching, cable routing, termination and testing.
- Implement appropriate network security controls and configuration practices to reduce operational and cyber risks.
- Integrate the new solution with relevant legacy ICT infrastructure with minimal disruption to hospital services.
- Procure required equipment, materials and services transparently and in accordance with applicable municipal procurement requirements.
- Train three hospital ICT staff members so they can operate, monitor and perform defined first-line support activities on the implemented solution.

- Complete commissioning, documentation, acceptance testing and operational handover.
- Establish a clear governance structure, reporting cadence, issue escalation path and accountability framework.

4. Major Deliverables

ID	Deliverable	Acceptance Description
D1	Project initiation package	Approved charter, stakeholder register, governance arrangements and baseline plan.
D2	Requirements & site assessment	Validated requirements, site survey, existing-network assessment and constraints register.
D3	Network & security design	High-level/low-level design, addressing/VLAN plan, security architecture and implementation plan.
D4	Procurement package	Specifications, quantities, quotations/tender documentation and procurement records as applicable.
D5	Civil and fibre works	Approved trenching, ducting/cabling, fibre termination, labelling and test results.
D6	Network implementation	Configured network equipment, connectivity, routing/switching and required integration.
D7	Security implementation	Security configuration, access controls, hardening, logging/monitoring and security verification.
D8	Integration & testing	Integration with relevant legacy services, test scripts, defect log and signed acceptance results.
D9	Training	Training delivered to three hospital ICT staff, attendance record and training materials.
D10	Handover & closure	As-built documentation, asset information, credentials handover process, support arrangements, lessons learned and closure report.

5. Project Scope

5.1 In Scope

- Project governance, planning, coordination, progress reporting and stakeholder management.
- Technical requirements gathering and validation.
- Site inspection and assessment of existing network infrastructure.
- Network architecture and implementation design.
- Fibre route planning, trenching coordination, cable installation and fibre termination.
- Network equipment configuration and connectivity implementation as defined by the approved design.
- Security design and implementation, including appropriate access controls and hardening.
- Legacy integration, configuration adjustments and controlled migration activities.
- Testing, commissioning, defect correction and acceptance.
- Training for three hospital ICT staff members.
- Technical documentation, as-built information and operational handover.

5.2 Out of Scope

- Major building works unrelated to the ICT/fibre route.
- Unapproved expansion of the network beyond the defined project sites or requirements.
- Replacement of unrelated legacy systems that are not required for the agreed integration.
- Long-term managed services beyond the agreed warranty/support period.
- New software/application development unless specifically approved through change control.
- Additional equipment, routes or sites requested after baseline approval without approved change control.

6. Project Organisation & Governance

Role	Primary Accountability	Decision/Authority Focus
Project Sponsor — Sekhukhune District Municipality	Provides strategic sponsorship, funding oversight and executive support.	Strategic approvals, escalation, major scope/budget decisions.
Project Manager —	Leads integrated delivery and coordinates all workstreams.	Day-to-day project decisions within delegated authority.

Network Engineer (NETENG)	Network design, configuration, connectivity, testing and technical documentation.	Network technical recommendations and implementation decisions.
Security Specialist (SEC)	Security architecture, hardening, access controls, security testing and risk advice.	Security controls and security acceptance recommendations.
Procurement Officer (PROC)	Procurement process, sourcing coordination and compliance documentation.	Procurement process decisions within delegated authority.
Subcontractor (SUB)	Trenching, fibre installation and termination activities.	Execution of contracted field works under approved method statements and supervision.
Hospital ICT Staff (3)	Requirements validation, training, legacy integration and operational readiness.	Operational acceptance and first-line support readiness.

Governance principle: the Project Manager coordinates the workstreams, while specialist roles retain technical/professional accountability within their areas. Material decisions affecting scope, budget, security, service continuity or contractual obligations must follow the agreed approval and change-control process.

7. High-Level Responsibility Matrix (RACI)

A = Accountable

R = Responsible

C = Consulted

I = Informed.

Activity	Sponsor	PM	NETENG	SEC	PROC	SUB / Hospital*
Project charter approval	A/R	C	C	C	I	I
Requirements & site assessment	A	R	R	C	C	R
Network design	A	R	R	I	C	C
Security design	A	C	R	I	I	C
Procurement process	A	R	C	R	C	I
Trenching & fibre works	A	R	C	I	I	R
Network implementation	A	R	R	I	C	C

Testing & acceptance	A	R	R	C	I	R
Training	A	R	C	I	C	R
Final handover	A/R	R	R	C	C	R

The final column represents the relevant operational participants depending on the activity. A = Accountable, R = Responsible, C = Consulted, I = Informed.

8. High-Level Requirements

ID	Area	Requirement
REQ-01	Connectivity	Provide stable connectivity between the defined project locations/endpoints.
REQ-02	Fibre	Install, terminate, label and test fibre according to the approved technical standard and route.
REQ-03	Network	Configure switching/routing and related services in accordance with the approved design.
REQ-04	Security	Apply least-privilege access, secure management, appropriate segmentation, hardening and logging.
REQ-05	Legacy integration	Integrate with required existing services while maintaining service continuity.
REQ-06	Performance	Verify agreed performance, availability and connectivity requirements through documented tests.
REQ-07	Documentation	Provide as-built diagrams, configuration records, test evidence and operational documentation.
REQ-08	Training	Train three hospital ICT staff members and verify basic operational competency.

9. Success Criteria & Acceptance

- All approved in-scope deliverables are completed and accepted.
- Fibre installation and termination pass applicable optical and physical tests, with test evidence retained.

- Network connectivity and configured services pass the approved test scripts.
- Security controls pass agreed verification checks with no unresolved critical/high-risk defects at go-live unless formally accepted by authorised management.
- Legacy integration is completed with documented rollback arrangements and no unacceptable disruption to critical services.
- Three hospital ICT staff members complete training and receive appropriate operational documentation.
- As-built network diagrams, asset information and relevant configuration/documentation are handed over.
- Outstanding defects are either closed or documented in an approved punch list with owners and due dates.
- Sponsor/project acceptance is formally recorded.

10. High-Level Project Schedule

Phase	Activity	Indicative Timing	Key Output
Phase 1	Initiation & mobilisation	Weeks 1–2	Charter, governance, stakeholders, initial plan
Phase 2	Requirements & site assessment	Weeks 2–4	Validated requirements, site survey, constraints
Phase 3	Design	Weeks 4–6	Network/security design and implementation plan
Phase 4	Procurement	Weeks 5–9	Sourcing, evaluation, approvals and orders
Phase 5	Civil/fibre works	Weeks 8–12	Trenching, ducting, fibre installation and termination
Phase 6	Network & security implementation	Weeks 11–14	Configuration, hardening and connectivity
Phase 7	Integration & testing	Weeks 13–15	Legacy integration, testing, remediation
Phase 8	Training & handover	Weeks 15–16	Training, documentation, acceptance and closure

The schedule is indicative and should be baselined after site assessment, procurement lead-time confirmation and approval of the detailed project plan.

11. Initial Risk Register

ID	Risk	Likelihood	Impact	Mitigation / Response
R-01	Procurement delays	Medium/High	High	Start specifications early; maintain procurement tracker; escalate approval bottlenecks.
R-02	Unknown underground services	Medium	High	Conduct site survey/service-location checks; approve method statement; use controlled excavation.
R-03	Fibre damage or poor termination	Low/Medium	High	Qualified technicians, installation standards, inspection and optical testing.
R-04	Legacy integration failure	Medium	High	Document dependencies; test in controlled windows; prepare rollback plan.
R-05	Cybersecurity misconfiguration	Medium	High	Security review, hardening checklist, least privilege and pre-go-live security verification.
R-06	Service interruption during cutover	Medium	High	Change window, communication plan, backup configuration and rollback procedure.
R-07	Scope creep	Medium	Medium	Formal change request, impact analysis and approval before implementation.
R-08	Subcontractor performance	Medium	Medium	Clear statement of work, milestones, quality inspections and

				progress reporting.
R-09	Budget overrun	Low/Medium	High	Baseline cost, contingency control and approval for variations.
R-10	Insufficient operational readiness	Medium	Medium	Training, handover checklist, knowledge transfer and support transition.

12. Assumptions

- Required project sites and technical areas will be accessible when needed.
- Relevant municipal/hospital stakeholders will be available for requirements validation, approvals and testing.
- Existing infrastructure information will be made available where reasonably obtainable.
- Procurement approvals and funding will be processed within agreed governance timelines.
- The subcontractor will provide appropriately skilled personnel and comply with site safety requirements.
- Required network/fibre equipment and materials will be available within confirmed lead times.
- Hospital ICT Staff (3) will be available for training and integration activities.

13. Constraints

- Hospital operations may restrict work windows and require careful change management.
- Budget and procurement rules may limit supplier selection and timing.
- Existing legacy technology may constrain design choices or migration sequencing.
- Civil works may be affected by site conditions, access, weather or unknown underground infrastructure.
- The project must maintain appropriate security and confidentiality throughout implementation.

14. Dependencies

- Approval of project charter and baseline scope.
- Availability of technical specifications and site information.

- Procurement approvals, supplier lead times and delivery of equipment/materials.
- Access permissions and coordination with affected facilities.
- Availability of hospital ICT staff for testing, training and operational sign-off.
- Completion of fibre/civil works before final connectivity commissioning.

15. Communication & Reporting Plan

Communication	Frequency	Owner	Audience	Content
Project status report	Weekly	Project Manager	Sponsor + project team	Progress, schedule, risks, issues, decisions
Technical workstream meeting	Weekly / as needed	Project Manager + NETENG	Technical team	Design, implementation, testing
Security review	At design, pre-go-live and as required	SEC	PM + NETENG + relevant stakeholders	Security risks, controls, exceptions
Procurement tracker	Weekly during procurement	PROC	PM + Sponsor	Sourcing, approvals, orders, lead times
Site coordination	Daily during field works	PM / SUB	NETENG + site representatives	Safety, progress, access, defects
Steering/escalation review	Monthly / milestone-based	Sponsor	PM + key stakeholders	Major decisions, escalations, changes

16. Change Control

Any request that changes approved scope, requirements, architecture, budget, schedule, security posture, site coverage or contractual deliverables must be documented as a change request before implementation. The Project Manager should assess impact on cost, time, resources, risks, quality, security and service continuity. Changes requiring sponsor or delegated authority approval must not proceed until the appropriate approval is recorded.

- Log change request and assign unique identifier.
- Describe reason, requested change and affected baseline items.
- Perform impact assessment.
- Obtain technical, security and procurement input where relevant.
- Obtain approval from the authorised decision-maker.
- Update scope, schedule, cost, risk and documentation baselines.
- Communicate approved change to affected stakeholders.
- Verify and close the change after implementation.

17. Quality Management

- Use approved technical designs and method statements before implementation.
- Inspect fibre routes, installation quality, labelling, termination and physical protection.
- Retain fibre test results and network test evidence as part of the project record.
- Apply configuration standards and peer review for material network/security changes.
- Maintain a defect/punch-list register through testing and handover.
- Do not declare operational acceptance until agreed acceptance criteria are satisfied or formally waived.

18. Cybersecurity & Information Protection

- Use least-privilege administrative access and secure management protocols.
- Change default credentials and protect privileged credentials using approved processes.
- Segment network services where required by the approved design.
- Harden network devices and disable unnecessary services/interfaces.
- Enable appropriate logging, monitoring and time synchronisation where supported.
- Control configuration backups and protect them from unauthorised access.
- Review security changes before production deployment.
- Record and formally approve any security exceptions.

19. Health, Safety & Environmental Considerations

- The subcontractor must comply with applicable site safety requirements and approved method statements.
- Trenching and excavation must be controlled to reduce risks to people, infrastructure and underground services.
- Work areas should be secured, signposted and restored appropriately after completion.
- Waste, packaging and removed materials should be handled in accordance with applicable environmental and site requirements.
- Hospital infection-control and access requirements must be considered where work occurs in or near clinical areas.

20. Handover & Operational Readiness

- Approved as-built network diagrams.
- Fibre route, termination and test documentation.
- Network or security configuration documentation appropriate for operational support.
- Asset or equipment register updates.
- Support and escalation contacts.

- Training materials and attendance evidence for three hospital ICT staff.
- Known issues, warranties and maintenance information.
- Final acceptance certificate.

21. Project Charter Approval

Approval of this charter confirms that the project is authorised to proceed into detailed planning and execution within the approved governance framework. It does not replace procurement approvals, technical design approvals, budget approvals, statutory/site permissions or other formal authorisations that may be required.

Approver Role	Name/Entity	Signature	Date
Project Sponsor	Sekhukhune District Municipality	Tlo~R	19 August 2026
Project Manager		Tlo~R	19 August 2026
Network Engineer	NETENG	Tlo~R	19 August 2026
Security Specialist	SEC	Tlo~R	19 August 2026
Procurement Officer	PROC	Tlo~R	19 August 2026

Project Organisation Diagram

PROJECT SPONSOR
Sekhukhune District Municipality

PROJECT MANAGER

WORKSTREAM LEADS
Network Engineer (NETENG) | Security Specialist (SEC) | Procurement Officer (PROC)

FIELD DELIVERY
Subcontractor (SUB) — Trenching & Fibre Termination

OPERATIONAL READINESS
Hospital ICT Staff (3) — Training & Legacy Integration

Field	Value
Document Title	Project Charter — ICT Network, Cybersecurity & Fibre Connectivity Project
Sponsor	Sekhukhune District Municipality
Project Manager	
Version	1.0
Status	Draft for Approval
Effective Date	Upon approval
Review Cycle	At major phase gates and whenever a material approved change affects the charter

REFERENCES

Gido, J., Clements, J. & Baker, R. (2018). *Successful Project Management*. 7th ed. Boston: Cengage Learning.

Gido, J., Clements, J.P. and Baker, R. (2018) *Successful Project Management*. 7th edn. Boston, MA: Cengage Learning, pp. 39–42, Chapter 2: 'Identifying and Selecting Projects' — section: 'Project Selection'. (The project was selected using the project selection process outlined by Gido, Clements and Baker (2018, pp. 39–42), which involves developing evaluation criteria, identifying assumptions, gathering relevant data and information, and evaluating each potential project against the established criteria.)

The authors organise proposals around technical, management and cost considerations (Gido, Clements & Baker, 2018, pp. 64–78).

The proposed network infrastructure project follows the project proposal principles presented by (Gido, Clements & Baker, 2018).